Customer Behavior Analysis Using E-Commerce Data

Advanced Analysis: Relationships and Behavioral Insights

Anshika Chauhan

Data Science / Data Analytics Project

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Objective

The goal of this phase is to move beyond basic descriptive statistics and identify deeper patterns in customer behavior. This includes analyzing relationships between user activity, transaction behavior, and revenue generation to uncover actionable insights.

Correlation Analysis

- To understand how different variables interact, correlation analysis was performed on key metrics such as purchase frequency, total revenue, and product pricing.
- A strong positive relationship was observed between purchase frequency and total revenue, indicating that highly active users contribute disproportionately to business growth. Additionally, a moderate correlation between product price and order value suggests that higher-priced items increase transaction size but may impact purchase frequency.
- These findings help in identifying key drivers of revenue and user engagement.

Key Behavioral Patterns

The dataset reveals several important behavioral trends:

- A small percentage of users contribute a significant portion of total revenue.
- Repeat users demonstrate higher consistency in purchasing behavior.
- Orders containing multiple items result in higher transaction values.
- Certain product categories dominate in terms of both demand and revenue contribution.

These patterns highlight opportunities for customer segmentation and targeted strategies.

Feature Engineering (New Metrics)

To enhance the analytical depth of the project, additional metrics were created:

CUSTOMER LIFETIME VALUE (CLV)

Measures total revenue generated by each user across all transactions.
→ Helps identify high-value customers.

AVERAGE ORDER VALUE (AOV)

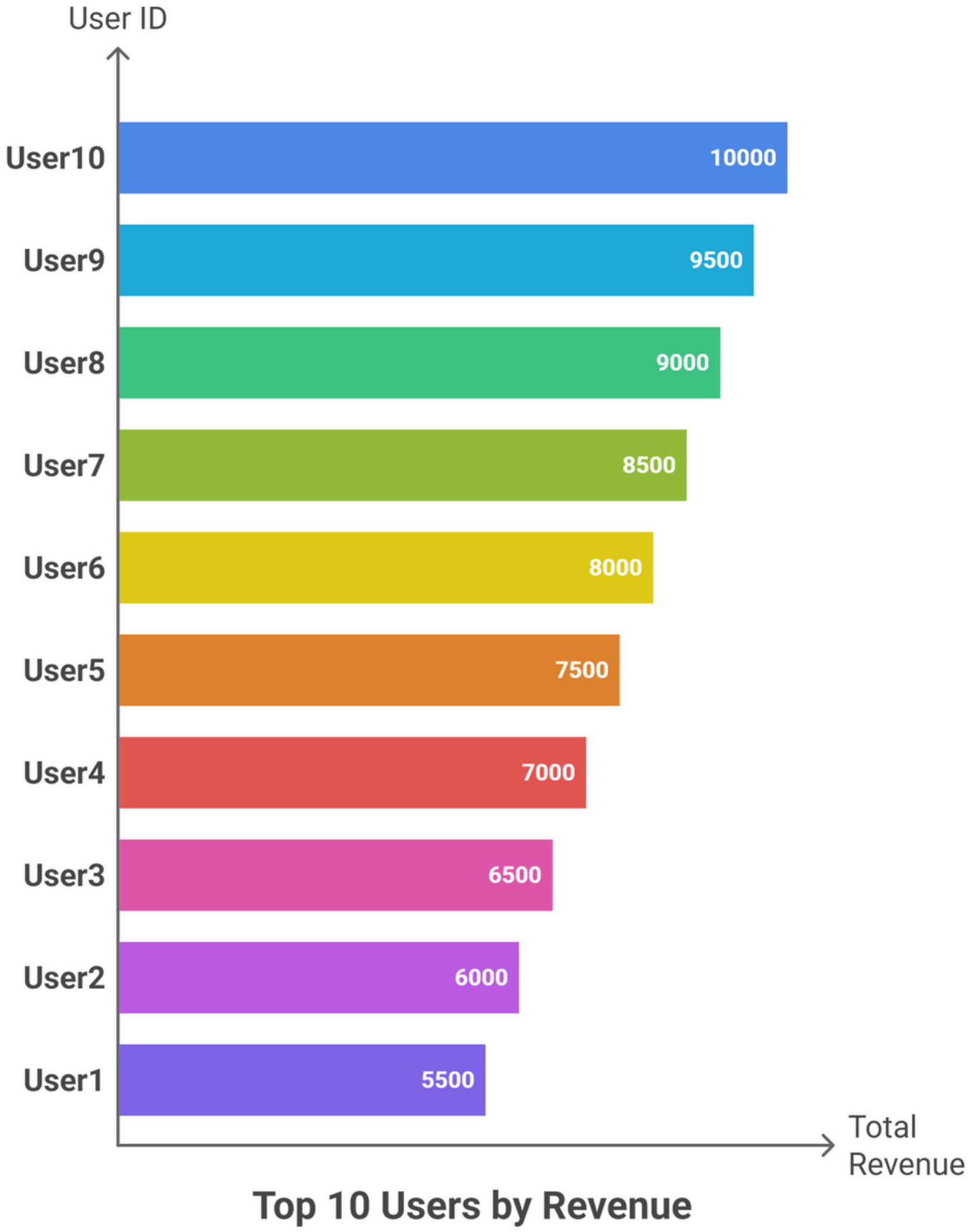
Calculated as total revenue divided by total number of orders.
→ Provides insight into spending behavior per transaction.

PURCHASE FREQUENCY

Number of orders placed per user.
→ Indicates customer engagement and retention.

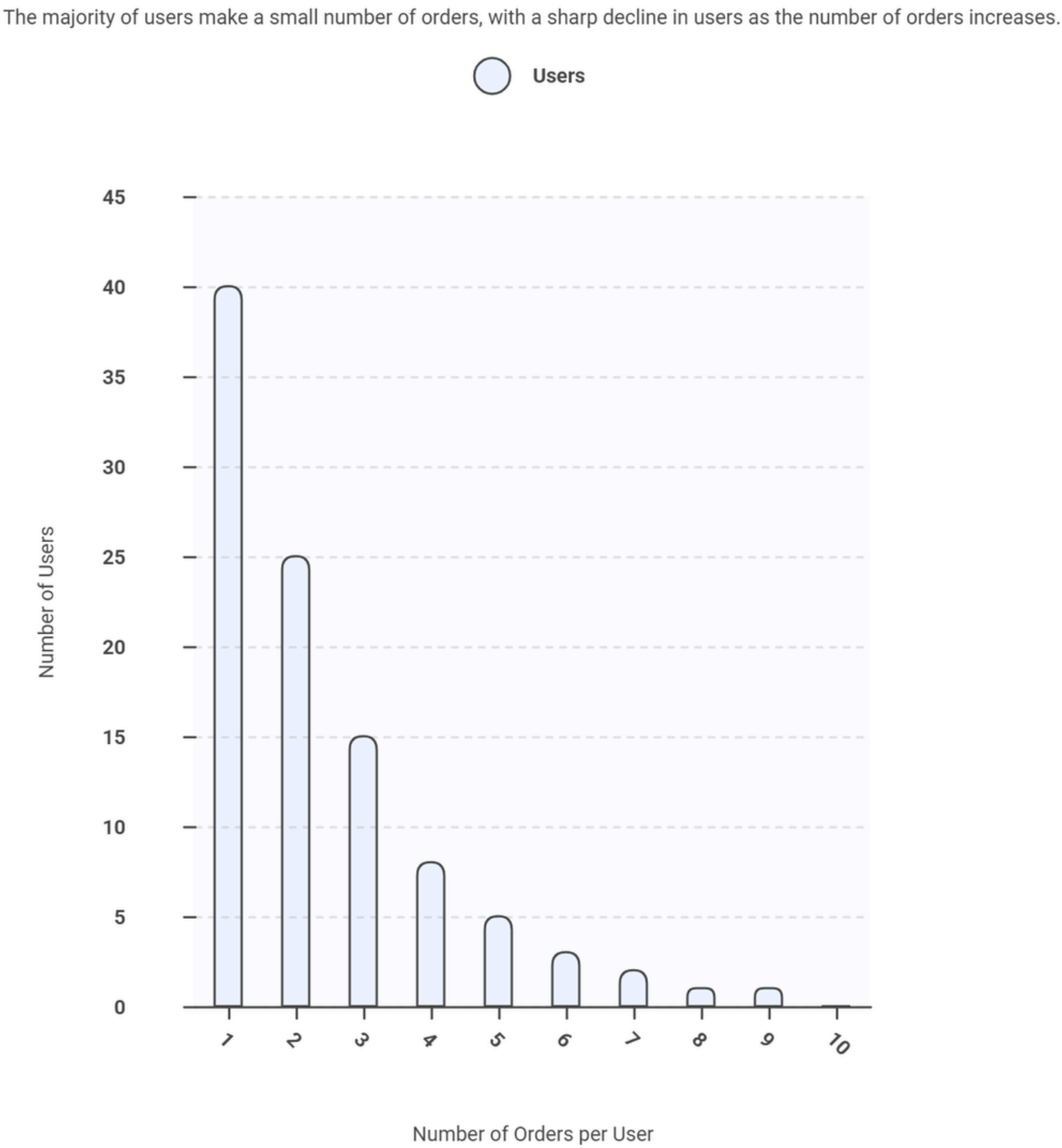
These engineered features allow more meaningful analysis compared to raw data alone.

Data Visualization: Revenue Distribution Among Users



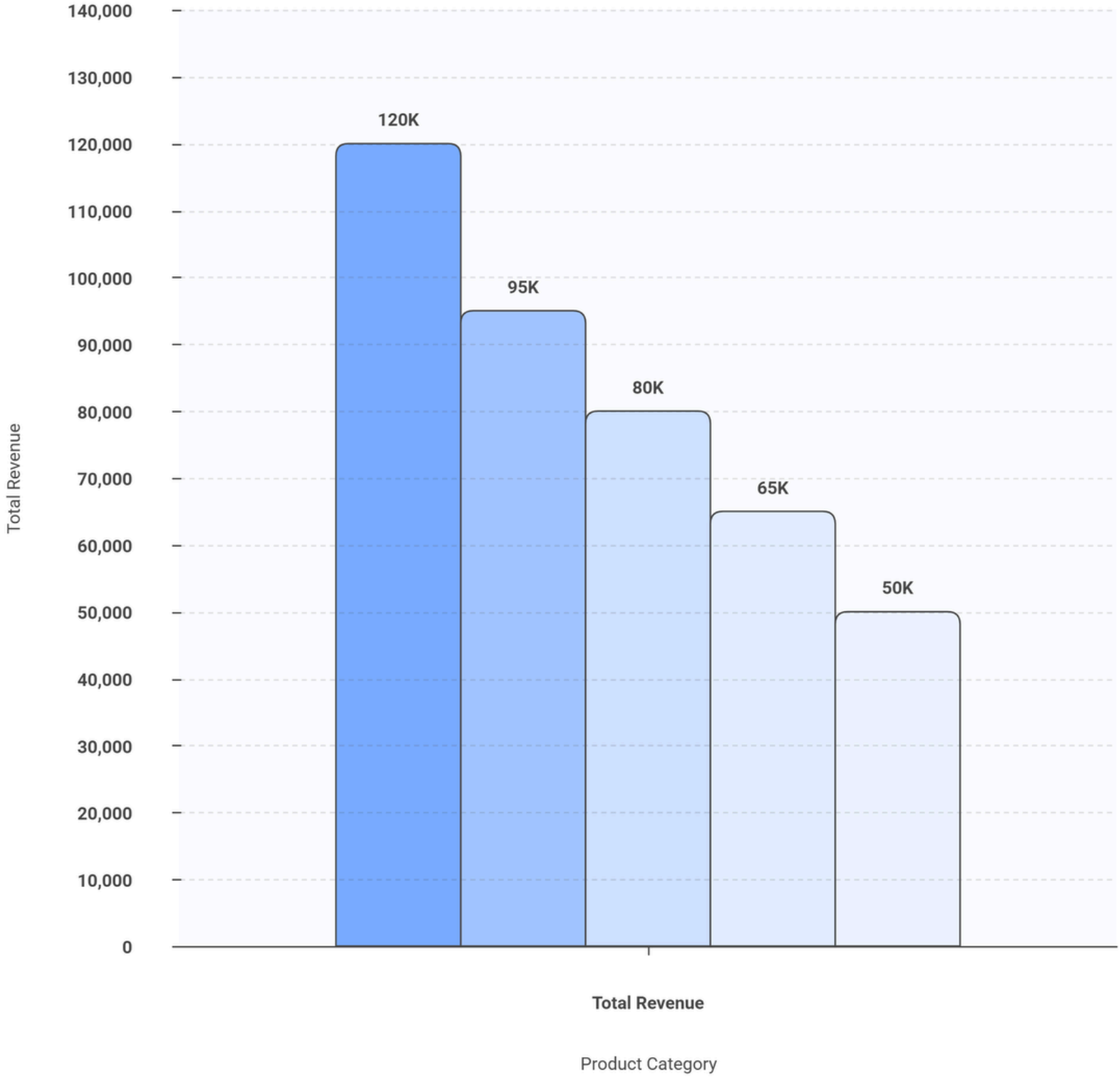
Insight:
This chart shows that revenue is concentrated among a small group of users, indicating that a limited number of high-value customers contribute significantly to overall business performance. This highlights the importance of focusing on retention strategies for top users.

Data Visualization: Purchase Frequency Distribution



Insight:
This chart shows that most users have a low purchase frequency, while a small group of users are highly active. This indicates that improving customer retention and encouraging repeat purchases can significantly impact overall engagement and revenue.

Data Visualization: Revenue by Product Category



Insight:
This chart shows that revenue is not evenly distributed across product categories. Electronics generates the highest revenue, followed by clothing and home-related products, while categories like sports contribute relatively less.
This indicates that certain product categories drive the majority of business revenue, highlighting opportunities for focused marketing, inventory planning, and category-level optimization.

Key Insights

- High-frequency users are the primary contributors to overall revenue.
- Revenue distribution is highly skewed, indicating dependency on a smaller user base.
- Product pricing influences transaction size but not necessarily purchase frequency.
- Repeat customers are more valuable than one-time buyers.

Business Interpretation

The analysis suggests that focusing on customer retention and repeat purchases can significantly improve revenue performance. Strategies such as personalized recommendations, loyalty programs, and targeted promotions can help increase engagement among high-value users.
Additionally, optimizing product pricing and identifying top-performing categories can further enhance profitability.

Next Steps

- Perform cohort analysis to study retention trends over time
- Build predictive models for customer lifetime value
- Analyze category-level performance in greater detail
- Implement segmentation for targeted marketing strategies

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